

Math 340: Elementary Matrix and Linear Algebra

MIDTERM TWO

Thursday, April 10th, 2025, 7:30pm-9:00pm

Circle your Instructor and your TA:

Dr. Lars Niedorf	Dr. Jose Rodriguez	Dr. Yassine Tissaoui	Dr. Ruhui Jin	Dr. Timur Yastrzhembskiy
Karthik Ravishankar	Kanav Madhura	Ankit Raheja	Joey Yu Luo	Yiyu Wang
Zaidan Wu	Dylan Jamner	Yijie He	Peter Wei	Tu Cao

Name: _____ Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of eight problems for a total of 50 points; the exam is six sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- If making multiple attempts at a solution, indicate clearly which attempt you'd like to be graded by crossing out the other answers. Multiple attempts will not be graded.
- When doing multiple choice and true/false questions, make sure to fully fill in the chosen bubble.
Correct mark: ● Incorrect marks: ● ⊗ ⊘ ⊙
- **Notation reminders:** M_{mn} is the vector space of $m \times n$ real matrices with standard matrix addition and scalar multiplication, P_d is the vector space of polynomials (with single variable x) of degree at most d with standard polynomial addition and scalar multiplication, $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.
- **Additional notation reminders:** O denotes the zero matrix, and I_n denotes the $n \times n$ identity matrix. $\text{tr}(A)$ denotes the trace of a matrix A .

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (7 points total) Suppose $T : \mathbb{R}^3 \rightarrow M_{22}$ is a linear transformation with $T \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$,

$$T \left(\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3 & 1 \\ 4 & 0 \end{bmatrix}, \text{ and } T \left(\begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix} \right) = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

a. (5 points) Compute $T \left(\begin{bmatrix} 4 \\ -5 \\ 2 \end{bmatrix} \right)$. Show all work. You do not need to simplify your final numerical answer.

b. (2 points) Is T onto? Justify your answer.

2. (6 points total) Let $A = \begin{bmatrix} 8 & 0 & -5 \\ -20 & -2 & 10 \\ 10 & 0 & -7 \end{bmatrix}$. You are given the following information about A : A has exactly two eigenvalues $\lambda = -2$ and $\lambda = 3$, and a basis for the eigenspace for $\lambda = -2$ is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$.

a. (4 points) Find a basis for the eigenspace of A for $\lambda = 3$. Show all work.

b. (2 points) Is A diagonalizable? Justify your answer.

3. (8 points, 1 point each) Clearly mark the correct answer for each of the following by **completely** filling in the appropriate bubble. **No justification is needed.**

a. **(True/False)** If U is a subspace of V and $\mathbf{u} + \mathbf{v}$ is in U , then $\mathbf{u}$ and $\mathbf{v}$ are both in U .

- ☐ True
☐ False

b. **(True/False)** Suppose A and B are invertible 2×2 matrices with $A + B = O$. Then $\{A, B\}$ is a linearly independent set.

- ☐ True
☐ False

c. **(True/False)** Suppose A and B are invertible 2×2 matrices with $A + B = O$. Then the columns of A is a linearly independent set, and the columns of B is a linearly independent set.

- ☐ True
☐ False

d. **(True/False)** The set of noninvertible 3×3 matrices is closed under scalar multiplication.

- ☐ True
☐ False

e. **(True/False)** Any set of 7 matrices in M_{23} must span M_{23} .

- ☐ True
☐ False

f. **(Multiple Choice-choose one)** Suppose A is similar to $B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 10 \end{bmatrix}$. Which of the following must be true about A ?

- ☐ A has trace 0.
- ☐ A is diagonalizable.
- ☐ The eigenvectors of A are the same as the eigenvectors of B .
- ☐ A is similar to I_3 .
- ☐ All of the above.

g. **(Multiple Choice-choose one)** Suppose $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis for a vector space V . Which of the following is also a basis for V ?

- ☐ $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}\}$
- ☐ $\{\mathbf{u}, \mathbf{u} + \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w}\}$
- ☐ $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}, \mathbf{v}\}$
- ☐ $\{\mathbf{u} - \mathbf{v}, \mathbf{u} + \mathbf{v}, \mathbf{w}\}$
- ☐ All of the above.
- ☐ None of the above.

h. **(Multiple Choice-choose one)** Which vector space has the same dimension as M_{24} ?

- ☐ $\mathbb{R}^6$
- ☐ P_8
- ☐ M_{42}
- ☐ All of the above.
- ☐ None of the above.

4. (6 points total) Consider P_2 and the following subset W of polynomials:

$$W = \{p(x) \text{ in } P_2 \mid p'(0) = 0\}.$$

(Reminder: $p'(x)$ is the derivative of $p(x)$.)

a. (2 point) Show that W is nonempty.

b. (2 points) Show that for $p_1(x), p_2(x)$ in W , $p_1(x) + p_2(x)$ is in W . Fully justify your answer.

c. (2 points) Show that for $p(x)$ in W and c in $\mathbb{R}$, $cp(x)$ is in W . Fully justify your answer.

6. (4 points) Consider the vector space P_1 and the set $S = \{2x + 10, c^2 - 41 - 5x\}$. Find all values of c so that the set S is linearly dependent.

7. (6 points total, 2 points each) Each of the following statements is **false**. Show each statement is false by providing explicit counterexamples. You must fully justify your answers.

a. Let V be $\mathbb{R}^2$ with the operation $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \oplus \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ y_1 + y_2 \end{bmatrix}$. Then $\oplus$ satisfies the vector space property $\mathbf{v} \oplus \mathbf{u} = \mathbf{u} \oplus \mathbf{v}$.

b. $F : M_{22} \rightarrow \mathbb{R}$ by $F(A) = \det(A) + \operatorname{tr}(A)$ is a linear transformation.

c. $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \text{ in } \mathbb{R}^2 : |x| \geq |y| \right\}$ is a subspace of $\mathbb{R}^2$.

8. (5 points total, 1 point each) Suppose A is a 3×5 matrix whose reduced row echelon form $RREF(A)$ is

$$RREF(A) = \begin{bmatrix} 1 & 3 & -2 & 0 & 4 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Answer each of the following questions about the matrix A , if possible. Justification is not required.

- a. **(Multiple Choice-choose one)** What is the $\dim(\text{null } A)$?

- ☐ 0
☐ 1
☐ 2
☐ 3
☐ 4
☐ Not enough information to answer.

- b. **(Multiple Choice-choose one)** Is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ in the column space of A ?

- ☐ Yes
☐ No
☐ Not enough information to answer.

- c. **(Multiple Choice-choose one)** For $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, what is the dimension of the image of T ?

- ☐ 0
☐ 1
☐ 2
☐ 3
☐ 4
☐ 5
☐ Not enough information to answer.

- d. **(Multiple Choice-choose one)** For $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, is T one-to-one (also called injective)?

- ☐ Yes
☐ No
☐ Not enough information to answer.

- e. **(Multiple Choice-choose one)** Are the (five) column vectors making up A a linearly independent set in $\mathbb{R}^3$?

- ☐ Yes
☐ No
☐ Not enough information to answer.

Scratch paper page!! (Must be turned in with exam)

Another Scratch paper page!! (Must be turned in with exam)